GLASS-BOX GREEN INSIGHTS

Dataset Build Report

What was built, which sources were used, what each file contains, and how the data was validated.

Status: all 33 automated validation checks passed

Branch

feature/dataset-pipeline

Dataset version 1.0.0 | Seed 42 | Cutoff 2026-06-30 | 90-day outcome window

1. Work completed

A complete, reproducible hybrid dataset pipeline was created for the four required Lloyds opportunities. It combines bounded public-source samples with synthetic banking data. No real or confidential bank data is used.

Output	Rows
Customers	5,000
Transactions	276,346
Customer products	16,937
Vehicles	3,416
Monthly energy records	54,060
Monthly digital records	57,684
Customer 360 records	5,000
Customer-product labels	20,000

The pipeline also creates transaction labels, product reference data, official emissions factors, customer-level splits, demonstration scenarios, validation reports, a data dictionary, checksums and an Excel summary.

REQUIREMENT MAPPING

2. Brief requirements covered

Requirement	Dataset implementation
Customer 360	One model-ready row per customer with transaction, product, vehicle, energy and digital features.
Eight transaction categories	Every transaction has exactly one gold label from the required mutually exclusive vocabulary.
Four products	One take-up label per customer for EV Finance, Green Home Loan, Solar Finance and Paperless Banking.
Cutoff and outcome window	Features end 2026-06-30. Positive labels occur only in the following 90 days.
Customer-level splits	3,500 train, 750 validation and 750 test customers with no overlap.
Documented labels	Latent factors, observable weak signals and random noise create plausible, non-circular labels.
Missing/conflicting data	Missing mileage, incomplete energy/digital history, existing EVs and ownership conflicts are included.
Documented factors	Reference table uses exact units and values from the revised 2026 UK government factor workbook.

ML predicts take-up. This dataset does not treat propensity as financial eligibility.

SOURCE PROVENANCE

3. Public datasets used

UCI Online Retail II

20,000 valid positive transactions were sampled deterministically from the official 1,067,371-record source. Customer IDs were hashed. Real item-description text and amount distributions seed 35,771 final OTHER-category transactions. Final customer IDs and dates remain synthetic.

https://archive.ics.uci.edu/dataset/502/online+retail+ii

London smart-meter data

The official partitioned archive is approximately 759 MB compressed and about 10 GB expanded. The downloader uses HTTP range requests to read 100,000 half-hour observations from one official partition, then retains 71 hashed household-month aggregates. Those profiles seed all 54,060 final energy rows.

https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d

UK Government GHG Conversion Factors 2026

The revised flat-format workbook supplies exact factors for petrol, diesel, natural gas, UK grid electricity and electricity transmission/distribution. Values remain in a separate versioned reference table.

https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2026

Public sources seed realistic patterns. They are not presented as Lloyds banking data.

4. Dataset structure

Public samples + synthetic archetypes → raw tables → Customer 360 + labels → customer splits → validation

Raw layer

- customers.csv: demographics, region, home ownership, preferences and archetype.
- transactions.csv: dated merchant text, amount, debit/credit, gold category and source basis.
- customer_products.csv: historical products plus post-cutoff synthetic take-up events.
- vehicles.csv: fuel type, age, mileage, efficiency and ownership status.
- energy_usage.csv: monthly electricity/gas usage, solar flag, meter type and quality.
- digital_usage.csv: monthly app/web usage, transactions, branch visits and statement preference.

Processed layer

- customer_360.csv: one row per customer with only pre-cutoff features.
- product_takeup_labels.csv: one binary outcome for every customer-product pair.
- transaction_category_labels.csv: one gold category for every transaction.
- demo_scenarios.csv: stable IDs for missing-data, conflict, existing-EV and paperless demonstrations.

Reference and governance

- green_products.csv defines the four products and intended use cases.
- emission_factors.csv records factor value, unit, scope, source year and version.
- data_dictionary.csv documents every generated column.
- file_inventory.csv stores row counts, sizes and SHA-256 hashes.

EXACT OUTPUTS

5. Generated file inventory

File	Rows	Columns	Size
data/external/samples/london_smart_meter_monthly_sample.csv	71	6	3.8 KB
data/external/samples/uci_online_retail_sample.csv	20,000	6	1703.1 KB
data/processed/customer_360.csv	5,000	50	1724.2 KB
data/processed/demo_scenarios.csv	4	2	0.1 KB
data/processed/product_takeup_labels.csv	20,000	7	1355.4 KB
data/processed/transaction_category_labels.csv	276,346	2	4943.8 KB
data/raw/customer_products.csv	16,937	8	1183.7 KB
data/raw/customers.csv	5,000	13	556.6 KB
data/raw/digital_usage.csv	57,684	10	2698.5 KB
data/raw/energy_usage.csv	54,060	9	4493.1 KB
data/raw/transactions.csv	276,346	8	22095.5 KB
data/raw/vehicles.csv	3,416	11	230.5 KB
data/reference/emission_factors.csv	5	9	1.0 KB
data/reference/green_products.csv	4	7	0.6 KB
data/reports/validation_checks.csv	32	3	2.0 KB
data/splits/test_customers.csv	750	2	8.8 KB
data/splits/train_customers.csv	3,500	2	44.5 KB
data/splits/validation_customers.csv	750	2	13.2 KB

6. Label generation

A customer is first assigned a hidden green-attitude value and a hidden financial-readiness value. Product labels combine those hidden values with several observable weak signals and random noise. The hidden values are not written to Customer 360.

Product	Signals used in the synthetic probability
EV Finance	Fuel spending, mileage, financial readiness, green attitude and noise.
Green Home Loan	Home ownership, electricity/gas use, financial readiness, green attitude and noise.
Solar Finance	Home ownership, electricity use, no existing solar, readiness, green attitude and noise.
Paperless Banking	Paper statements, mobile engagement, email engagement and noise.

Observed take-up rates

Product	Rate
EV_FINANCE	9.8%
GREEN_HOME_LOAN	11.9%
PAPERLESS_BANKING	32.7%
SOLAR_FINANCE	11.2%

These are simulated labels. They are suitable for pipeline development and comparative validation, not for claims about real customer demand.

7. Leakage protection

- All feature transactions are dated on or before 2026-06-30.
- Energy and digital histories also end before the cutoff.
- Take-up dates exist only after cutoff and within the 90-day outcome window.
- Future green-product rows are filtered before Customer 360 is built.
- Customer 360 contains no take_up or take_up_date fields.
- All records for one customer remain in exactly one of train, validation or test.
- The final test split is held separately so modelling code can avoid tuning on it.

Split	Customers	Share
Train	3,500	70%
Validation	750	15%
Test	750	15%

Validation confirmed zero customer overlap between splits.

AUTOMATED CHECKS

8. Validation results

Overall result: PASS. 32 checks passed.

Check	Status	Evidence
customers_unique	PASS	5,000 customer rows
transactions_unique	PASS	276,346 transaction rows
vehicles_unique	PASS	3,416 vehicle rows
customer_360_one_per_customer	PASS	5,000 feature rows
transactions_foreign_keys	PASS	unknown customer IDs: 0
products_foreign_keys	PASS	unknown customer IDs: 0
vehicles_foreign_keys	PASS	unknown customer IDs: 0
energy_foreign_keys	PASS	unknown customer IDs: 0
digital_foreign_keys	PASS	unknown customer IDs: 0
labels_foreign_keys	PASS	unknown customer IDs: 0
transaction_categories_allowed	PASS	['AUTOMOTIVE', 'ELECTRICITY', 'FUEL', 'GAS', 'HOME_IMPROVEMENT', 'OTHER', 'PUBLIC_TRANSPORT', 'UTILITIES']
all_categories_present	PASS	{'OTHER': 149028, 'UTILITIES': 23217, 'FUEL': 22433, 'PUBLIC_TRANSPORT': 20913, 'HOME_IMPROVEMENT': 17297, 'AUTOMOTIVE': 15821, 'ELECTRICITY': 15784, 'GAS': 11853}
one_transaction_label_each	PASS	276,346 labels
transaction_amounts_positive	PASS	range 1.0 to 1157.2
feature_transactions_before_cutoff	PASS	latest 2026-06-30
energy_before_cutoff	PASS	latest 2026-06-01
digital_before_cutoff	PASS	latest 2026-06-01
one_label_per_customer_product	PASS	20,000 of 20,000 expected
label_products_allowed	PASS	['EV_FINANCE', 'GREEN_HOME_LOAN', 'PAPERLESS_BANKING', 'SOLAR_FINANCE']
takeup_dates_in_outcome_window	PASS	positive labels: 3,283
negative_labels_have_no_date	PASS	no future date for negative labels
no_label_columns_in_features	PASS	forbidden columns found: []
future_products_excluded_from_features	PASS	future green-product flags absent
customer_splits_disjoint	PASS	overlap: 0
customer_splits_complete	PASS	covered: 5,000 of 5,000
split_ratio	PASS	{'train': 3500, 'validation': 750, 'test': 750}
energy_nonnegative	PASS	energy values are nonnegative
digital_nonnegative	PASS	digital counts are nonnegative
public_uci_seed_used	PASS	seeded rows: 35,771
public_london_seed_used	PASS	seeded rows: 54,060
missing_data_scenarios_present	PASS	customers below complete: 2,089
existing_ev_scenario_present	PASS	existing EV customers: 524

WHAT MUST NOT BE CLAIMED

9. Important limitations

- The customer population, banking relationships and product take-up are synthetic.
- UCI Online Retail II is retail invoice data, not merchant-level bank transaction data.
- The London data is a bounded source slice and is used for profile seeding, not population inference.
- Historical smart-meter readings are scaled into synthetic 2025-2026 profiles.
- Take-up model performance will describe synthetic-label performance only.
- Age, income and region distributions are illustrative rather than Lloyds-calibrated.
- Eligibility remains a separate rule process and is not encoded into propensity labels.
- Paperless adoption is not counted as CO2e reduction without a documented lifecycle method.

Use the dataset for prototype development, evaluation design and demonstration—not real lending decisions.

10. How to rebuild and use it

Run the complete pipeline

```
PYTHON_BIN=python3 bash scripts/build_data.sh
```

Run deterministic tests

```
python3 -m unittest -v tests.test_dataset
```

Main files for the ML owner

- data/processed/customer_360.csv
- data/processed/product_takeup_labels.csv
- data/raw/transactions.csv
- data/processed/transaction_category_labels.csv
- data/splits/*.csv

Main files for rules/carbon/integration

- data/raw/vehicles.csv and data/raw/customer_products.csv
- data/raw/energy_usage.csv and data/raw/digital_usage.csv
- data/reference/green_products.csv
- data/reference/emission_factors.csv
- data/processed/demo_scenarios.csv

Review artifacts

- data/reports/validation_report.json contains machine-readable checks.
- data/reports/dataset_summary.xlsx contains reader-friendly summaries, sources and inventory.
- data/data_dictionary.csv describes all generated columns.

The pipeline is ready for model development on feature/dataset-pipeline.

FINAL STATUS

Dataset built.
Sources documented.
Leakage checks passed.
Team handoff ready.

Version 1.0.0 | 33/33 validation checks passed

The official Lloyds problem statement remains the source of truth.